

to their capacity. Hence, we had to select between servers that were all loaded. In such a scenario, the *Ordering* property did not seem to hold in a few instances (about 1 in 5). However, this scenario encompasses only a very small portion of the overall problem space. Hence, we conclude that *pMapper* is able to make use of the *Ordering* property to quickly come up with placements that are very close to optimal (less than 0.2% penalty).

6 Related Work and Conclusion

Energy management in server clusters has been a popular area of research since the beginning of this decade [1,12]. Chen et al. [6] combine server level CPU scaling techniques with the application provisioning problem in the same formulation. However, in contrast to *pMapper*, they do not work in a virtualized setting with migration costs and only deal with homogeneous clusters for placement. Muse pose a resource allocation problem in [4], where services are allocated to enough number of resource containers on physical servers based on an economic model. Since energy cost is part of the model and the load is dynamic, resources are allocated dynamically in a manner that is aware of the energy costs. However, the model does not explicitly consider migration costs of containers or deal with the complexity of application specific power models in a heterogeneous server cluster. Bobroff et al. [2] describe a runtime application placement and migration algorithm in a virtualized environment. The focus is mainly on dynamic consolidation utilizing the variability in workload but they do not perform power-aware placements on heterogeneous servers.

Most of the cluster energy management literature addresses the problem of distributing requests in a web server cluster in such a way that the performance goals are met and the energy consumption is minimized [4,5,10,19,21]. There are a number of papers that describe server or cluster level energy management using independent [8,18] or cooperative DVS techniques [7,11]. There are other efforts in reducing peak power requirements at server and rack level by doing dynamic budget allocation among sub-systems [9] or blades [20].

In this work, we have presented an application placement controller *pMapper* that minimizes power and migration costs, while meeting the performance guarantees. *pMapper* differs from all existing literature because it addresses the problem of power and migration cost aware application placement in heterogeneous server clusters that support virtualization with live VM migration. It investigates the viability of using CPU utilization based application specific power models to develop placement algorithms and validates the assumptions through testbed experimentation. Through a carefully designed experimental methodology on two server platforms, we concluded that only a power minimization framework is feasible practically. We use insights from our study to pin down the conditions under which this problem can be solved. We proposed three dynamic placement algorithms to minimize power and migration cost and experimentally demonstrated the various scenarios in which each algorithm is effective. We established